SHIKHAR PANDAV

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PROFESSIONAL EXPERIENCE

Infosys Noida, India
Software Engineer *July 2024 – April 2026*

- **Engineered** distributed backend microservices in **Python (FastAPI)** serving **5,000+ requests per second (RPS)**, cutting **p95 API latency 35%** via **Redis** caching on hot read paths and database connection-pool tuning.
- **Built** automated **CI/CD** pipelines and **Prometheus** + Grafana observability, sustaining **99.99% uptime** and reducing production-incident **MTTR 30%** through automated alerting and faster triage.
- **Migrated** legacy REST endpoints to **gRPC** and **Protobuf**, shrinking network payload size **40%** and cutting inter-service error rates **25%**.
- **Re-architected** monolithic processes into a fault-tolerant, event-driven system using **RabbitMQ** and **Redis**, boosting high-frequency data-ingestion throughput **20%**.
- **Profiled GCP** services with **cProfile** and eliminated critical memory leaks in long-running workers, saving **$24,000 annually** in infrastructure costs.

TECHNICAL PROJECTS

CodeQuery AI – Distributed RAG Intelligence System
Backend Developer | Python, FastAPI, ChromaDB, Celery, Tree-sitter *January 2026 – February 2026*

- **Built** a high-performance **Retrieval-Augmented Generation (RAG)** engine using **Tree-sitter** for semantic, syntax-aware code chunking, increasing context-retrieval precision **45%** over fixed-window chunking.
- **Integrated ChromaDB** vector search with **cross-encoder re-ranking**, improving top-k relevance scores across a database of **100+** repositories.
- **Architected** asynchronous **Celery** + **Redis** task queues to offload heavy AI inference, eliminating API blocking during large-scale indexing. GitHub

MyTube – Scalable Full-Stack Video Infrastructure
Full Stack Developer | Next.js, PostgreSQL, Drizzle, Mux *May 2025 – July 2025*

- **Engineered** an end-to-end video streaming pipeline with the **Mux API** for adaptive-bitrate streaming, supporting **10,000+** concurrent viewers with sub-2s buffering.
- **Accelerated** relational query performance **50%** using **PostgreSQL** and **Drizzle ORM**, leveraging **Common Table Expressions (CTEs)** for complex nested retrieval. GitHub | Website

TECHNICAL SKILLS

Languages: Python, C++, SQL (PostgreSQL/MySQL), JavaScript (Node.js/React), C#
Backend & DevOps: Microservices, FastAPI, Django, Docker, GCP, RabbitMQ, gRPC, Redis, CI/CD
AI & Data: RAG, Vector Databases (Pinecone, ChromaDB), LangChain, Tree-sitter, NumPy, Pandas
Development Tools: Git, Celery, Linux, REST APIs, Unit Testing (Pytest), System Design

ACHIEVEMENTS

- **LeetCode Guardian**: Peak Rating **2212** (Top 1% globally); Solved **1,500+** problems on Data Structures and Algorithms.
- **Codeforces Specialist** (Peak Rating **1429**).
- **Global Rank 430** / **21,482** in LeetCode Weekly Contest.

EDUCATION

KIET Group of Institutions, AKTU Delhi-NCR, India
Bachelor of Technology in Computer Science and Engineering *August 2020 – June 2024*